

FRAME DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-421 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-422 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

INSTRUMENT DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

CALIBRATE DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MODEL DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-425 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

VALIDATE DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-426 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

EVALUATE DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-427 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

OPTIMIZE DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-428 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

STRESS-TEST DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-429 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-430 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.